

# Jesse Clark

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Permanently eligible to work in Australia and in the United States.

## Summary

Product-oriented, research-minded computer scientist.

Obsessed with cool technology like single-molecule sensors, social robots, and spaceflight.

- Developed product prototypes and demonstrations used to secure venture capital and grant funding.
- Produced visualizations and automated reports to help find insights in data for space mission teams, B2C sales departments, and scientific instruments.
- Improved productivity and reliability of data-driven research labs.
- Experience with regulatory review for spaceflight and medical devices.

## Employment

**Geneus Technologies**, Mountain View, California

May 2023 – May 2024

### *Principal Algorithm Engineer*

Established dry-lab operations at a new office for a Chinese biotechnology company (深圳今是科技有限公司).

- Hired a data science team, managed local projects, and reported to executives overseas.
- Developed primary analysis algorithms for a nanopore DNA sequencing instrument.
- Designed statistical models to infer hidden variables from raw time-series sensor data, and calculate Phred quality scores.
- Selected high-accuracy algorithms to produce neural network training data, and high-performance algorithms for use in production.

Python C++ PyTorch Machine Learning Data Science Signal Processing State Estimation Time Series Analysis DNA Sequencing

**Roche Molecular Diagnostics**, Santa Clara, California

December 2019 – May 2023

### *Principal Software Engineer*

Developed software for [nanopore DNA sequencing](#) instruments. Adapted mature algorithms for the needs of new experiments, including molecule detection, level-calling, base-calling, quality metrics, and neural network training.

Coordinated data compatibility between wet-lab experimentalists, GPU-accelerated signal processing, machine learning model training, and statistical simulations. Created validation tools to enforce data compatibility agreements and prevent unnecessary processing or troubleshooting.

Python TypeScript C++ TensorFlow Machine Learning AirFlow Jenkins Data Science DNA Sequencing

**University of Technology, Sydney**, Ultimo, New South Wales

November 2014 – December 2019

### *Senior Software Engineer*

Responsible for software operations for [The Innovation and Enterprise Research Lab](#).

- We won first place in the global [RoboCup@Home](#) competition. I was responsible for software system reliability, sequential task planning, and natural language processing.
- Implemented robot behaviors for experiments and demonstrations. Presented research results at conferences.
- Developed curriculum and delivered university lectures and masterclasses.

Python JavaScript iOS Swift SVG CI/CD Keras Machine Learning ROS Human-Robot Interaction Teaching

**Sydney Programming School**, Mosman, New South Wales

April 2015 – December 2019

### *Head of Teaching*

Delivered programming lessons at an after-school program, as it grew from 0 to over 1,000 students.

Scratch JavaScript Arduino Python Processing Lego Mindstorms Robotics Cryptography Blockchain Game Design Teaching

## Modus Ponens, LLC

Founded a new business to develop web and mobile applications for Silicon Valley startups. Led teams of local and remote designers and developers. Had contact with prominent venture capitalists.

- **Maverick Poker** 2014
- **OneLessDesk** 2014
- **Destination Software** 2013 – 2014
- **GoBoldly** 2013
- **Scanadu** 2012
- **SolarCity** 2012
- **Soundwave** 2011
- **Robot Panda Productions** 2010 – 2011
- **WoWGen** 2008

Node.JS JavaScript HTML CSS REST Progressive Enhancement Riak Erlang iOS Objective-C C# Android Java Ruby on Rails  
Parse.com Firebase.com Salesforce.com Distributed Systems GIS Augmented Reality

## NASA Ames Research Center, Moffett Field, California

July 2007 – April 2011

### *Computer Scientist*

Developed applications for space mission workflows for the Space Shuttle and International Space Station.

Perl MySQL Lucene HTML JavaScript LaTeX Human-Computer Interaction Contextual Inquiry Root Cause Analysis

## NASA Goddard Space Flight Center, Greenbelt, Maryland

June 2004 – June 2007

### *Computer Engineer*

Developed robotics and graphics software to simulate contact dynamics for the Hubble Space Telescope.

Recovered and visualized 1960s climate data from **Nimbus** and **TIROS** weather satellite archives.

C++ OpenGL Perl Mathematica FANUC Robotics GNC

## Wheels of Zeus, Los Gatos, California

2003

### *Intern*

PIC Assembly Electrical Engineering Linux Apache HTTPD

## Institute for Computer Assisted Orthopaedic Surgery, Pittsburgh, Pennsylvania

May 2002 – July 2002

### *Intern*

C++ Perl Tcl/Tk Electrical Engineering OpenGL Augmented Reality

## Board Membership

### Johnson and Louise H. Clark Charitable Foundation, Saratoga, California

April 2023 – Present

#### *Treasurer*

Managing financial accounts for a family-run charitable foundation.

## Publications

See [curriculum vitae](#) for details.

GitHub link: <https://github.com/jjclark1982>

ResearchGate link: <https://www.researchgate.net/scientific-contributions/Jesse-Clark-2116266501>

## Education

### Carnegie Mellon University, Pittsburgh, Pennsylvania

May 2004

#### *Bachelor of Science in Computer Science, Minor in Mathematical Sciences*

(4-year degree with research component — equivalent to an Honours degree)

C++ Java Perl Mathematica Lisp OCaml x86 Assembly Prolog LaTeX Machine Learning Kernel Programming TCP/IP  
Digital Logic Cryptography Linear Algebra Cellular Automata Bioinformatics